Report on *"Matrix Theory Reloaded: A BPS Road to Holography"*

by C. D. A. Blair, J. Lahnsteiner, N. A. Obers and Z. Yan

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Summary of the paper

The paper treats the decoupling limit that produces matrix theory on Dp-branes as a *BPS decoupling limit of the whole of type II string theory*, not just of the brane. The resulting corner, "matrix p-brane theory" (MpT), has a non-Lorentzian target space with a codimension-$(p+1)$ foliation. It is described by p-brane Newton–Cartan vielbeins and has a p-brane Galilean boost symmetry. The paper then develops three strands.

(i) *Holography.* The paper applies the MpT limit at asymptotic infinity of a Dq-brane supergravity solution. For $p = q$ this limit *induces* the near-horizon (Maldacena/IMSY) limit in the bulk. For $p \neq q$ it gives a non-Lorentzian bulk geometry in which a second, MqT-type limit produces a near-horizon geometry. This reproduces the recent non-Lorentzian proposals of Lambert et al. in a systematic way.

(ii) *DLCQ conjecture.* Lifted to M-theory, each BPS limit corresponds to a DLCQ, up to U-duality. This motivates the conjecture that holography is a DLCQ^n/DLCQ^m correspondence with $m > n$. In this language AdS$_5$/CFT$_4$ is DLCQ^0/DLCQ^1, the non-Lorentzian examples are DLCQ^1/DLCQ^2, and D1–D5 AdS$_3$/CFT$_2$ is DLCQ^0/DLCQ^2.

(iii) $T\bar{T}$. Undoing the BPS limit is identified with the $T\bar{T}$ deformation for the string, and with U-dual "p-brane $T\bar{T}$" deformations for branes. This yields explicit flow equations:

- a closed-form flow in any dimension, taking D free scalars to the Dirac–Nambu–Goto action;
- a strikingly simple cubic flow for $p = 2$;
- flows including the gauge field for $p = 1, 2$;
- perturbative flows for $p \geq 3$.

1. Principal achievement

The main achievement is conceptual unification: the *near-horizon limit is not an independent ingredient of holography.* It is the bulk image of the BPS decoupling limit that defines the boundary theory, and the near-horizon geometry automatically approaches the flat non-Lorentzian MpT geometry at its own asymptotic boundary. Matrix theory, AdS/CFT and its IMSY generalisations, and the recently proposed non-Lorentzian holographic dualities were until now treated with separate, partly ad hoc scalings. Here they become entries of a single duality-organised table, generated by one rule applied repeatedly. The analysis also explains *why* only certain brane combinations work: $q = p + 4$ for aligned configurations, and $p + q = 10$ in general, unless one smears or rescales N. The bookkeeping in terms of iterated DLCQs gives a common vocabulary for flat-space, non-Lorentzian and (via timelike T-duality) Carrollian corners.

The most portable concrete result is the family of generalised $T\bar{T}$ flow equations,

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \frac{1}{2\lambda^2}\Big\{\mathrm{tr}(\mathbb{1} - \lambda\mathcal{T}) - (p-1)\big[\det(\mathbb{1} - \lambda\mathcal{T})\big]^{\frac{1}{p-1}} - 2\Big\}.$$

This equation reproduces the known $p = 0$ (Gross et al.) and $p = 1$ (standard $T\bar{T}$) cases. For $p = 2$ it gives a new and simple cubic flow. Unlike earlier higher-dimensional proposals, it reaches the full multi-scalar Nambu–Goto action, and it explains why ordinary $T\bar{T}$ acting on 2D Maxwell theory does not reach DBI.

I checked the equation numerically. Along the claimed solution family, the closed-form Nambu–Goto-type Lagrangian, both sides agree to numerical precision for $p = 0, 2, 3, 4, 5$ with 1, 3 and 6 scalars. The $p = 2$ polynomial form agrees as well. The claims are therefore correct as stated.

2. Principal drawback

The principal drawback is that the framework is *kinematic*: it reorganises limits and geometries, but no new duality or new deformation is put to a *dynamical* test. Several features of the paper show this.

- *The established dualities are re-described, not extended.* For $p = q$ the asymptotic MpT limit is shown (Sec. 3.1) to coincide with the Maldacena limit after a reparametrisation.
- *The new non-Lorentzian bulk backgrounds are not established.*
  - They are not shown to solve the equations of motion and torsion constraints of MpT supergravity, which the authors acknowledge.
  - Their isometries and supersymmetries are not computed.
  - Generic cases need smearing, or a by-hand ω-dependent rescaling of ℓ (that is, of N), whose physical meaning is left open.
- *No observable is matched on both sides.* No spectrum, symmetry algebra, thermodynamic quantity, protected correlator or index is computed for any proposed new duality. The DLCQ^n/DLCQ^m conjecture therefore works as a classification scheme without a criterion that could falsify it. Its DLCQ^2/DLCQ^3 example (Spin Matrix Theory) rests on an admittedly tentative identification of the third DLCQ.
- *The $T\bar{T}$ "intrinsic generation" of the bulk is partly circular.*
  - The radial profile $\Omega(r)$ is inserted by hand using the dilatation symmetry; the deformation does not determine it.
  - The p-brane flows are classical, bosonic and abelian.
  - They are fixed "by construction" only along the DBI family. The operator is not unique off this family: in the single-scalar case the paper's flow coincides with the inequivalent flow of Ferko et al.
  - There is no quantum definition, and no analogue of Zamolodchikov factorisation.

Above all, the step that matters for holography is missing: applying the deformation to the large-N, non-abelian worldvolume theory. As a result, the paper's answer to its own Question 3 is currently a statement about a single classical brane action, not about the strongly coupled theory dual to the bulk. A secondary issue is length. The paper runs to roughly 100 pages, much of it review, and this dilutes the genuinely new content.

3. Most worthwhile follow-up

Proposal: a large-N thermodynamic test of "p-brane $T\bar{T}$ = undoing the near-horizon limit", starting with the M2-brane ($p = 2$). *Why this case.* The $p = 2$ flow is exact, polynomial and independent of the field strength. The paper derives it for both D2- and M2-branes, and M-theory has no running dilaton. It is therefore the cleanest place to test the claim that the deformation interpolates between the near-horizon $AdS_4 \times S^7$ geometry and the full, asymptotically flat brane solution (the "1" in $H = 1 + L^6/r^6$ restored).

Plan.

(a) *Field-theory side.* Apply the cubic flow to ABJM on T^2 at large N. Use large-N factorisation of the composite operators $\mathrm{tr}\,\mathcal{T}^2$, $(\mathrm{tr}\,\mathcal{T})^2$ and $\det\mathcal{T}$, following the treatment of Taylor's operator by Hartman, Kruthoff, Shaghoulian and Tajdini. This gives a closed, Burgers-type equation for $E(\lambda; S, V)$ in homogeneous states, which can be integrated from the conformal equation of state $\varepsilon \propto s^{3/2}$.

(b) *Gravity side.* Compute the thermodynamics of non-extremal M2-branes in asymptotically flat space at fixed charge, with λ identified through ω^{-2} and the paper's dictionary. One striking feature makes the test promising. At fixed charge, $T = 6/(4\pi r_0 \cosh\beta)$ is bounded and has two branches: a near-extremal branch that reduces to AdS_4 thermodynamics, and an unstable far-from-extremal branch. This mirrors the maximal (Hagedorn-like) temperature and square-root branch structure of $T\bar{T}$ with the corresponding sign.

(c) *Compare* the free energies $F(T, \lambda)$, including the location of $T_{\max}$. Then extend the comparison to:

- the D3-brane, using the scalar flow plus the perturbative F-terms;
- the D1–D5 system with its double deformation. This is S-dual to the single-trace $T\bar{T}$ of Giveon–Itzhaki–Kutasov, where an analogous match is known to work.

Feasibility and impact. Both sides are classical or large-N computations with established tools, so the project is realistic on a time-scale of months. Either outcome is informative.

- A match would give the first dynamical evidence for the paper's central claim. It would also supply a higher-dimensional, $T\bar{T}$-like deformation with a concrete asymptotically flat holographic dual, which matters for flat-space holography.
- A mismatch would show whether the deformation must be implemented in a "single-trace" rather than a factorised ("double-trace") way. It would thereby constrain the missing non-abelian completion of the flow.

The main risks are two. Factorisation of the cubic $\det\mathcal{T}$ term in $d = 3$ needs justification, and the map between λ and the supergravity parameters must be fixed with care. Neither looks prohibitive. The project addresses the paper's principal weakness directly, at modest cost.